

Oluwatimilehin Emmanuel Owolabi

Also appears as Timilehin or Timi

Lagos, Nigeria | timilehin.owolabi@stu.cu.edu.ng | +2349027257724 | [Website](#) | [Scholar](#) | [LinkedIn](#) | [GitHub](#) | [ORCID](#)

Education

Covenant University

B.Eng. in Electrical Engineering

Ogun, Nigeria

September 2020 – August 2025

- GPA: 4.85/5.00

- Thesis: [Development of Quadrotors with Adaptive Control for Autonomous Swarm Operations](#)

Research Interests

Self-improving embodied AI; meta-reinforcement learning; robot foundation models; hybrid learning and model-based control; mobility and manipulation for legged and aerial robots; stability and safety in data driven systems.

Research Experience

University of Oxford — Dr. Daniel Omeiza, Oxford Robotics Institute

Oxford, United Kingdom (Remote)

Undergraduate Research Assistant

March 2024 – September 2024

- Analyzed the Lyft Level 5 dataset to estimate multi-agent vehicle positions and velocities, detect lane-change maneuvers, and improve explainability of autonomous driving decisions in complex traffic scenarios using both ego- and agent-centric representations.
- Implemented subset-scan algorithm on Graph Neural Networks to identify statistically significant anomalies in datasets.

Google Developer Groups on Campus

Ogun, Nigeria

Robotics Research Team Lead

September 2024 – Present

- Trained a vision-based manipulation policy using DeepMind Control Suite, MuJoCo Playground, and JAX, achieving reliable box-picking performance in simulation.
- Prototyped a low-cost 3D-printed manipulator for object sorting, matching the performance of the WidowX 250 S at just 2.5% of its cost; concurrently supervised mechanical, electrical, and software teams.
- Led the replication of the ALOHA paper in simulation, using OpenVLA-OFT to teach students bimanual manipulation with vision-language-action models.
- Mentored 16 students on critical review and implementation of robotics research papers, deepening their understanding and improving replication accuracy.

Covenant University Centre for Research, Innovation and Discovery

Ogun, Nigeria

Research Student

September 2024 – Present

- Assisted with doctoral, master's, and undergraduate research projects on artificial intelligence and robotics, including an autonomous garbage-picking robot and a machine learning model for pandemic prediction in African countries.
- Developed an NLP program to parse, summarize, and cluster Scopus alert emails, improving research discoverability.

Zummit Africa

Lagos, Nigeria

Machine Learning Research Intern

April 2023 – May 2024

- Led a brain tumor segmentation project, achieving 95% accuracy using U-Net and medical imaging datasets.

Peer-Reviewed Publications (Journals and Conferences)

A. Awelewa, N. K. Ojo, T. Abimbola-Oladejo, **T. Owolabi**, A. Saha and V. Srivastava, "Fuzzy-PID Controller for Liquid Level Control of Tank Systems", Accepted for presentation at the IEEE International Conference on Science, Engineering and Business for Driving Sustainable Development Goals, 2025.

T. E. Somefun, **T. Owolabi**, and O. M. Longe, "[Practical Trade-offs in Neural Network Optimization: Brute Force Search and Gradient Descent](#)", Engineering Research Express, 2025.

O. Idowu, H. Hu, A. Akinwale, A. Ilori, Z. Xingze, Y. Wang, A. Rasheed, and **T. Owolabi**, "[Enhancing Radiological Imaging for Better Healthcare Outcomes Through High Performance Hybrid Approach](#)", ASRIC Journal on Engineering Sciences, 2024.

T. E. Somefun, N. Elijah, A. E. Igbo-Orere, **T. Owolabi**, C. Somefun and S. Ongbali, "[Energy Optimization Algorithm for Reducing Energy Consumption in a Smart Home](#)", International Conference on Multidisciplinary Engineering and Applied Science (ICMEAS), 2023.

Teaching Experience

Association of Professional Women Engineers of Nigeria	Lagos, Nigeria
Machine Learning Tutor	April 2024 – Present
• Delivered in-depth machine learning classes to 27 female graduate and undergraduate students.	
Project Cage Technologies	Lagos, Nigeria
Robotics Instructor	May 2024 – July 2024
• Led the senior robotics team to the finals of the Pan African Robotics Competition Stars League 2024.	

Awards and Honors

Finalist, Pan-African Robotics Competition	2024
Third Place, DSN Covenant University Hackathon (\$30)	2022
Founder’s Award, Best WASSCE Result, Wellspring College (\$50)	2021
Valedictorian, Department of Science, Wellspring College (\$100)	2020
Salutatorian, Overall Graduating Class, Wellspring College (\$100)	2020
Lux Award, Most Intellectual Student, Wellspring College	2020

Selected Projects

Autonomous Delivery Quadcopter	Ongoing
• Designed and constructed a custom ESP32-based flight controller hardware and software.	
Action Recognition and Tracking System	2023
• Developed an action-recognition and tracking system using a hybrid CNN-LSTM model for monitoring and logging patient actions, detecting falls, and issuing emergency alerts, integrated with a PID controller for subject tracking.	

Selected Presentations and Invited Talks

AI: Our Current Reality and Future Trajectory
Nigerian Society of Engineers (NSE), August 2024
Deep Learning: Neural Networks
Association of Professional Women Engineers of Nigeria (APWEN), April 2024

Selected Certifications

Deep Learning Specialization Certificate	Coursera / DeepLearning.AI
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Core Skills

Technical Skills: Python, C++ , MATLAB, ROS2, PyTorch, Tensorflow, MuJoCo, JAX, Raspberry Pi, STM32, ESP32
Personal Skills: Leadership, Hyper-focus, Hardwork, Resilience, Growth Mindset, Teamwork

Professional Experience

Scrella	Lagos, Nigeria
Lead Machine Learning Engineer	July 2024 – Present
• Designed and deployed machine learning operations pipelines to optimize workflow efficiency.	
• Used Vision Language Models for confidential video analysis, improving workers decision speed.	
Peak Fiction	Belfast, Northern Ireland (Remote)
Machine Learning Intern	April 2024 – July 2024
• Utilized Latent Diffusion Models to create high-quality comic book scenes.	
• Engineered an Image Generation workflow using ComfyUI, increasing productivity by 50%.	
Schneider Electric	Lagos, Nigeria
Sustainability Intern	August 2023 – October 2023
• Conducted robotics workshops for young students on building a simple self-driving car.	
• Collaborated with African schools and NGOs for STEM, contributing to a goal of empowering 1 million youths globally.	
• Assisted in analyzing and approving NGO support budgets, each totaling over \$60,000.	

Leadership Experience

Association of Electrical and Information Engineering Students	Ogun, Nigeria
President	September 2024 – Present
• Led a 20-member executive team in coordinating projects and competitions for over 2,000 students.	
• Launched a student mentorship program pairing juniors with seniors for academic and career guidance.	